A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Atmospheric pollution: smoke, acid rain, NOx, hydrocarbons, CO, smogs - Set 2

Mark scheme

AQA spec 3.4.3.2.1-3.4.3.2.5; Pollution; textbook pp. 245-254

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (52 - 40) / 40 x 100 **1** = 30% **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 3 points below. **1** Sulfur dioxide came mainly from burning coal – domestic coal fires were controlled by the Clean Air Acts and coal-fired power stations have closed or been fitted with flue gas desulfurisation **1** Low-sulfur fuels are now used, and sulfur can be removed from the fuel before combustion **1** Nitrogen oxides are produced whenever any fuel is burned in air at high temperature, so they are not removed by changing the fuel – road traffic in the city centre has continued to grow **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **flue gas desulfurisation** → removing sulfur dioxide from flue gases with lime or limestone	3
01.3	**1 mark each** 1 mark for each of the 3 points below. **1 Electrostatic precipitator:** the flue gases pass charged wires that give the particulates an electric charge, so they are attracted to oppositely charged collecting plates and can be shaken off and removed **1 Catalytic converter:** a platinum / rhodium catalyst converts oxides of nitrogen, carbon monoxide and unburnt hydrocarbons into nitrogen, carbon dioxide and water **1 Low-NOx burner:** the fuel is burned at a lower temperature, or in stages, so less of the nitrogen and oxygen in the air combines to form oxides of nitrogen **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Electrostatic precipitator** → charged plates that attract and collect particulates • **particulates** → fine solid particles / smoke and dust • **Catalytic converter** → a catalyst that converts NOx, CO and unburnt hydrocarbons to nitrogen, carbon dioxide and water	3
01.4	**1 mark each** 1 mark for each of the 3 points below. **1** Ground-level ozone is a secondary pollutant: it is not emitted directly but formed in the air by sunlight acting on oxides of nitrogen and volatile organic compounds **1** Nitrogen dioxide is split by ultraviolet light to give nitrogen monoxide and an oxygen atom, which joins an oxygen molecule to make ozone **1** With less NOx there is less of the raw material for these photochemical reactions, so less ozone and fewer PANs are produced **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • ground-level ozone described as protecting us from UV - that is stratospheric ozone; at ground level it is a pollutant	3

Do not write outside the box

Question	Marking guidance	Marks
01.5	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Lichens differ in their sensitivity to sulfur dioxide, so the species present show the average pollution level over a long period rather than at one instant **1** They need no power supply or expensive equipment and can be surveyed at many sites, giving good spatial coverage cheaply **1** They integrate the effects of pollution over years, so a short pollution episode may not show up **1** They indicate sulfur dioxide well but say little about nitrogen dioxide, ozone or particulates, which are now the city's main problems **1** Other factors affect them too, such as the pH and texture of the bark or stone they grow on, shade and humidity, so a survey can be misleading unless these are standardised **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **indicator species** → a species whose presence or absence shows the conditions present • **particulates** → fine solid particles / smoke and dust **Do not credit** • 'lichens die where the air is polluted' without naming the pollutant - it is their sensitivity to sulfur dioxide that is used	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 245-254; AQA spec 3.4.3.2.1-3.4.3.2.5.	

Do not write outside the box